

Lab 1: Understanding the Basic Network Equipment

Subject: Computer Network | Class: BSc.CSIT 4th Semester

Theory

A computer network is a system of interconnected computing devices that can communicate and share resources with each other. Networks rely on both hardware and software components to function. The hardware components — often called network equipment or network devices — are physical devices that connect computers, manage traffic, and ensure data reaches its correct destination. Understanding these devices is fundamental to designing, building, and troubleshooting any network.

A. Network Equipment

Network equipment refers to the physical hardware devices used to connect computers and other devices in a network. These devices control the flow of data, extend the range of networks, provide security, and allow communication between different types of networks. The major network equipment includes: Repeater, Hub, Switch, Bridge, Router, Modem, Network Interface Card (NIC), Firewall, Wireless Access Point (WAP), and VoIP Endpoint.

B. Repeater

i. Definition

A repeater is a network device that operates at the Physical Layer (Layer 1) of the OSI model. It receives a weak or degraded signal and retransmits it at a higher power level, allowing signals to travel longer distances without loss of quality.

ii. Function

The primary function of a repeater is to extend the transmission distance of a network signal. When data travels over a cable, the signal gradually weakens due to attenuation. The repeater amplifies and regenerates this signal so it can continue further along the network medium. It does not interpret or filter data — it simply boosts and retransmits the signal. Repeaters are used in both wired (e.g., Ethernet) and wireless networks.

iii. Figure

(Refer to Figure: Two computers connected through a Repeater — weak signal on one side, amplified signal on the other)

C. Hub

i. Definition

A hub is a basic networking device that operates at the Physical Layer (Layer 1) of the OSI model. It connects multiple computers or network devices in a Local Area Network (LAN) and serves as a common connection point for all devices in the network segment.

ii. Function

When a hub receives a data packet from one port, it broadcasts (copies) that packet to all other ports, regardless of the intended destination. This means all devices connected to the hub receive the data, and each device must determine whether the packet was meant for it. Hubs do not filter data or manage traffic intelligently. They create a single shared collision domain, which can lead to network congestion and reduced performance. Hubs are largely obsolete today, replaced by switches.

iii. Figure

(Refer to Figure: A HUB at the center with one SENDER and multiple RECEIVERS connected to it, showing broadcast to all ports)

D. Switch

i. Definition

A switch is a network device that operates at the Data Link Layer (Layer 2) of the OSI model. It connects multiple devices in a LAN and intelligently forwards data only to the specific device for which it is intended, using MAC (Media Access Control) addresses.

ii. Function

A switch maintains a MAC address table that maps each device's MAC address to its corresponding port. When it receives a data frame, it reads the destination MAC address and forwards the frame only to the port connected to the destination device. This reduces unnecessary traffic and eliminates collisions, giving each device its own dedicated communication channel. Switches significantly improve network performance and efficiency compared to hubs. They are the most common device used to build LANs today.

iii. Figure

(Refer to Figure: PC 'A' sends a 'Message for F' → Switch → Message forwarded only to PC 'F', not to B, C, D, or E)

E. Bridge

i. Definition

A bridge is a network device that operates at the Data Link Layer (Layer 2) of the OSI model. It is used to connect two or more network segments and controls data flow between them based on MAC addresses.

ii. Function

A bridge divides a large network into smaller segments to reduce traffic and collisions. It learns the MAC addresses of devices on each segment and maintains a filtering table. When a data frame arrives, the bridge checks whether the destination MAC address is on the same segment as the source. If it is, the bridge blocks the frame (filtering); if the destination is on a different segment, the bridge forwards it. This helps isolate network traffic and improves overall network efficiency.

iii. Figure

(Refer to Figure: BRIDGE connecting SEGMENT 1 (PC1, PC2 and other devices) to SEGMENT 2 (other devices), filtering traffic between segments)

F. Router

i. Definition

A router is a network device that operates at the Network Layer (Layer 3) of the OSI model. It connects multiple networks together and directs data packets from one network to another using IP addresses.

ii. Function

A router reads the destination IP address of each incoming packet and determines the best path for it to reach its destination using routing tables and routing protocols (e.g., OSPF, RIP, BGP). Routers connect different networks — such as a home LAN to the internet — and can connect networks using different technologies. They also perform Network Address Translation (NAT), which allows multiple devices to

share a single public IP address. Routers provide logical separation between networks and are essential for internet connectivity.

iii. Figure

(Refer to Figure: Internet → Modem → Router → Computer #1 and Computer #2)

G. Modem

i. Definition

A modem (short for Modulator-Demodulator) is a device that converts digital signals from a computer into analog signals suitable for transmission over telephone lines, and vice versa. It operates at the Physical Layer (Layer 1) of the OSI model.

ii. Function

When data is sent from a computer, the modem modulates the digital signal into an analog signal that can travel over the telephone network or cable infrastructure. At the receiving end, another modem demodulates the analog signal back into a digital signal for the receiving computer. Modems are used to provide internet access via DSL, cable, or fiber connections. They act as the bridge between a local network and the Internet Service Provider (ISP).

iii. Figure

(Refer to Figure: Laptop → Modem (Modulate digital signals) → Telephone Network (analog signals) → Modem (Demodulate analog signals) → Laptop)

H. Network Interface Card (NIC)

i. Definition

A Network Interface Card (NIC), also called a network adapter, is a hardware component installed in a computer or device that enables it to connect to a network. It operates at both the Physical Layer (Layer 1) and Data Link Layer (Layer 2) of the OSI model.

ii. Function

The NIC provides the physical interface between a computer and the network medium (cable or wireless). It converts data from the computer's internal format (parallel) into a format suitable for transmission over the network (serial), and vice versa. Each NIC is assigned a unique 48-bit MAC (Media Access Control) address by the manufacturer, which is used for device identification within a network. NICs can be wired (Ethernet) or wireless (Wi-Fi).

iii. Figure

(Refer to Figure: Physical image of a NIC — a green PCB circuit board with an RJ-45 Ethernet port and PCI connector)

I. Firewall

i. Definition

A firewall is a network security device (hardware or software) that monitors and controls incoming and outgoing network traffic based on predetermined security rules. It acts as a barrier between a trusted internal network and untrusted external networks such as the internet.

ii. Function

A firewall inspects data packets and allows or blocks them based on rules defined by the network administrator. It can filter traffic by IP address, port number, protocol, and content. Firewalls protect networks from unauthorized access, malicious attacks, viruses, and intrusions. They can be implemented as dedicated hardware appliances, software applications, or a combination of both. Modern firewalls also offer stateful inspection, deep packet inspection (DPI), and application-layer filtering.

iii. Figure

(Refer to Figure: Internet → Firewall → Internal network with User 1, User 2, and User 3)

J. Wireless Access Point (WAP)

i. Definition

A Wireless Access Point (WAP) is a networking device that allows wireless devices to connect to a wired network using Wi-Fi. It operates at the Data Link Layer (Layer 2) and acts as a central transmitter and receiver of wireless radio signals.

ii. Function

A WAP connects to a router or switch via an Ethernet cable and creates a wireless local area network (WLAN) in its coverage area. Wireless devices such as laptops, smartphones, and tablets connect to the WAP using Wi-Fi, gaining access to the wired network and the internet. WAPs are used in homes, offices, schools, and public spaces to extend wireless coverage. Unlike a router, a WAP does not route traffic — it simply provides the wireless interface between devices and the existing wired network.

iii. Figure

(Refer to Figure: ISP → Modem → Router → Switch → Wireless Access Point → Laptops (wireless); also Router → Laptops (wired))

K. VoIP Endpoint

i. Definition

A VoIP (Voice over Internet Protocol) Endpoint is a device or application that enables voice communication over an IP network (such as the internet) instead of the traditional Public Switched Telephone Network (PSTN). Common examples include IP phones, softphones (software-based phones), and computers with VoIP applications.

ii. Function

A VoIP endpoint converts analog voice signals into digital data packets, which are then transmitted over an IP network. On the receiving end, the device reassembles the packets and converts them back into audio. VoIP technology uses protocols such as SIP (Session Initiation Protocol) and RTP (Real-time Transport Protocol) for call setup and media transmission. VoIP endpoints allow for cost-effective voice, video, and multimedia communication over the internet, eliminating the need for traditional telephone infrastructure.

iii. Figure

(Refer to Figure: IP Device Caller → Voice Audio Breaks Up into Packets → Packets Sent Over Internet → Packets Reassemble in Order and Played Back as Audio → Call Destination)

Conclusion

This lab provided a comprehensive understanding of the fundamental network equipment used in modern computer networks. Each device plays a specific and important role in building functional, efficient, and secure networks. The Repeater and Hub operate at the Physical Layer and are responsible for signal amplification and broadcasting, while the Switch and Bridge work at the Data Link Layer to intelligently forward data using MAC addresses. The Router, operating at the Network Layer, connects different networks and routes packets using IP addresses. The Modem enables internet connectivity by converting digital signals to analog and back, while the NIC provides every computer with a unique hardware identity for network communication. The Firewall secures the network by filtering unwanted or malicious traffic, and the Wireless Access Point extends network connectivity to wireless devices. Finally, VoIP Endpoints enable voice and multimedia communication over IP networks, replacing traditional telephony. Understanding these devices forms the foundation for advanced topics in networking such as network design, IP addressing, routing protocols, and network security.

References

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